☑ CTL Provisional DECISION SHEET (PDSH)
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Standard(s) (incl. year)	Subclause(s)	Tracking No.
IEC 61010-1:2010 IEC 61010-1:2010/AMD1:2016	clause 6.5.4	2184A
Category		
MEAS		
Subject	**Keywords**	**Developed by**
Y1 capacitor bringing reinforced insulation	Protective impedance Y1 capacitor Impulse rating	ETF 3

Question

Clause 6.5.4 of the IEC 61010-1:2010 + A1:2016 requires the protective impedance to be one or more of the following:

a) an appropriate single component which shall be constructed, selected and tested so that safety and reliability for protection against electric shock is assured. In particular, the component shall be:

1) rated for twice the maximum working voltage

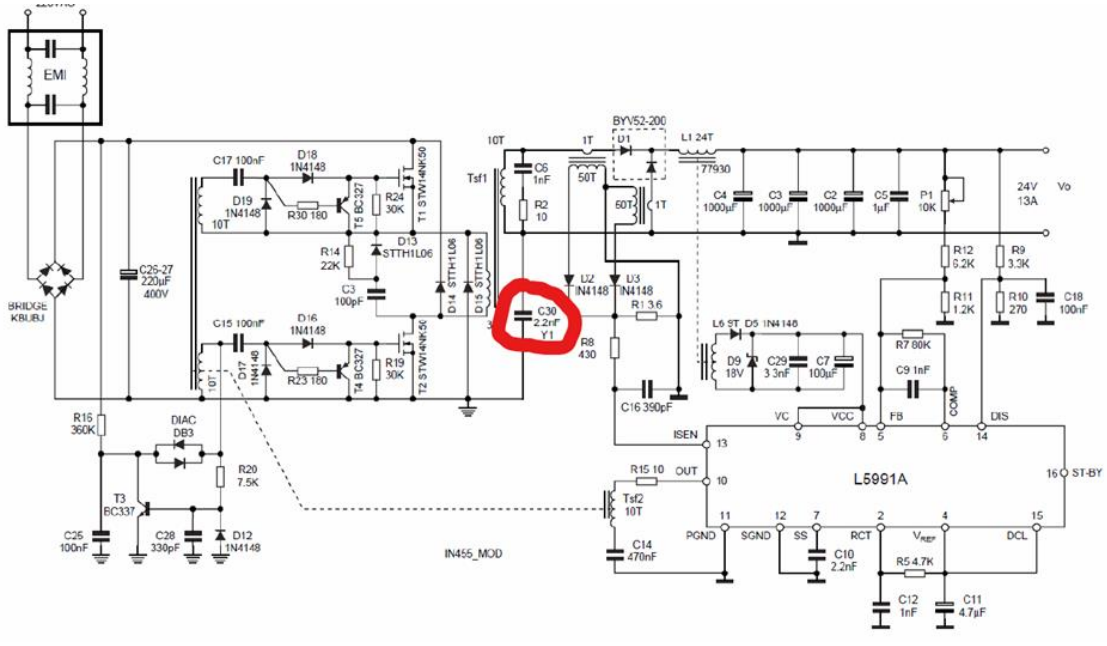
2) if a resistor, rated for twice the power dissipation for the maximum working voltage.

b) a combination of components.

In the case of a switch mode power supply, a capacitor is generally set between primary and secondary pins of a high frequency transformer to reduce the EMC radio frequency interference.

In the sample circuit, if the working voltage measured across the Y1 capacitor is 348 Vrms, the standard IEC 61010-1: 2010+A1: 2016 requires:

- 1x Y1 capacitor rated at 700Vrms (>2x348Vrms), or
- 2x Y1 capacitor rated at least to 400Vrms (>1x348Vrms)



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Question: Is a capacitor bridging a DI/RI (Double insulation / Reinforced insulation) barrier subject to the requirements of Clause 6.5.4?
Decision
No, clause 6.5.4 is not considered when reviewing a capacitor bridging insulation. The capacitor requirements are covered by Clause 14.1 of IEC 61010-1:2010 + A1:2016 and require the capacitor to comply with IEC 60384-14. For insulation requiring DI/RI (Double insulation / Reinforced insulation), the capacitor would need to be a subclass Y1 capacitor rated for the required working voltage and transient as defined in Clause 6.7 of IEC 61010-1: 2010+A1:2016. This approach aligns with and allows the use of IEC 62368 and IEC 60950 power supplies with IEC 61010-1:2010 + A1:2016 equipment.
Explanatory notes
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DSH Publication date:
Electrical engineering defines impedance as the combined effect of resistance and reactance that opposes alternating current. Reactance is made up of inductance and capacitance. This definition does not directly apply to the clause 6.5.4 use of the term impedance. In the meaning of IEC 61010-1 the resistance component is what forms protective impedance. Class Y capacitors (subclass Y1 or Y2) provide isolation by Double Insulation or Reinforced Insulation between terminals, and they do not rely on impedance.